

CHICAGO CALUMET HARBOR, IL, USA

WMO: 997255

Lat: 41.730N

Lon: 87.538W

Elev: 589

StdP: 14.39

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.4	(c) 5.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 25.0	(k) 26.1	(l) 22.8	(m) 30.8	(n) 9.4	(o) 250	(p) 0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.8	(c) 89.6	(d) N/A	(e) 86.9	(f) N/A	(g) 84.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 12.0	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 26.1	(b) 22.1	(c) 18.9	(e) -8.6	(f) 92.5	(g) 7.3	(h) 3.5	(i) -13.8	(j) 94.9	(k) -18.1	(l) 97.0	(m) -22.1	(n) 98.9	(o) -27.4	(p) 101.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	25.5	27.0	36.7	48.6	58.8	68.8	73.9	73.0	66.2	54.4	42.0	31.5
	DBStd	19.24	11.65	10.29	9.31	8.69	9.24	7.68	6.18	4.71	6.98	8.47	9.75	10.24
	HDD50	2881	762	644	428	128	21	0	0	0	0	48	274	576
	HDD65	6105	1226	1063	879	497	242	50	6	2	66	345	691	1038
	CDD50	3118	1	1	14	88	293	563	739	714	485	184	33	3
	CDD65	869	0	0	1	6	50	163	280	251	101	17	0	0
	CDH74	5726	0	0	2	47	399	1160	2058	1396	580	84	0	0
	CDH80	1701	0	0	0	3	113	344	702	368	157	14	0	0
(14) Wind	WSAvg	9.1	9.5	9.5	11.1	11.0	9.3	8.6	7.7	7.4	8.0	8.5	9.1	9.1
(15) Precipitation	PrecAvg	35.4	2.1	1.8	2.2	3.3	4.1	3.9	3.9	3.9	2.8	3.0	2.6	1.9
	PrecMax	48.1	5.1	5.9	4.9	6.9	7.5	10.9	9.7	8.8	9.4	8.5	6.7	4.3
	PrecMin	20.5	0.6	0.5	0.6	1.3	0.7	0.9	1.2	0.9	0.7	0.9	0.4	0.3
	PrecStd	6.3	1.2	1.1	1.1	1.4	1.7	2.0	1.9	2.0	1.9	1.9	1.6	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.6	58.0	71.6	79.8	88.0	91.5	95.5	90.7	89.6	82.0	69.5	60.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	51.3	50.3	63.8	74.4	83.2	87.5	90.5	87.7	84.4	75.9	64.1	54.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	45.7	44.1	57.2	69.5	78.7	84.0	87.1	84.3	80.4	71.2	60.1	49.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	40.5	39.3	50.0	64.3	73.6	80.6	84.0	81.1	76.7	67.1	56.0	44.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	11.8	11.7	14.0	14.5	13.2	11.8	10.7	11.9	12.8	12.2	10.9
		MCDBR	17.4	17.4	23.2	24.1	21.7	18.6	16.7	16.1	17.5	18.1	16.9	15.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.306	0.329	0.353	0.409	0.438	0.443	0.455	0.452	0.422	0.365	0.323	0.306
	taud		2.438	2.359	2.368	2.238	2.211	2.248	2.214	2.219	2.264	2.425	2.496	2.483
	Ebn at Noon		268	277	283	273	267	265	260	257	256	261	259	258
	Edh at Noon		24	31	35	43	45	43	44	43	38	29	23	22
(44) All-Sky Solar Radiation	RadAvg		506	776	1135	1463	1667	1918	1934	1680	1389	916	581	422
	RadStd		55	102	84	120	142	124	92	98	117	97	62	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page